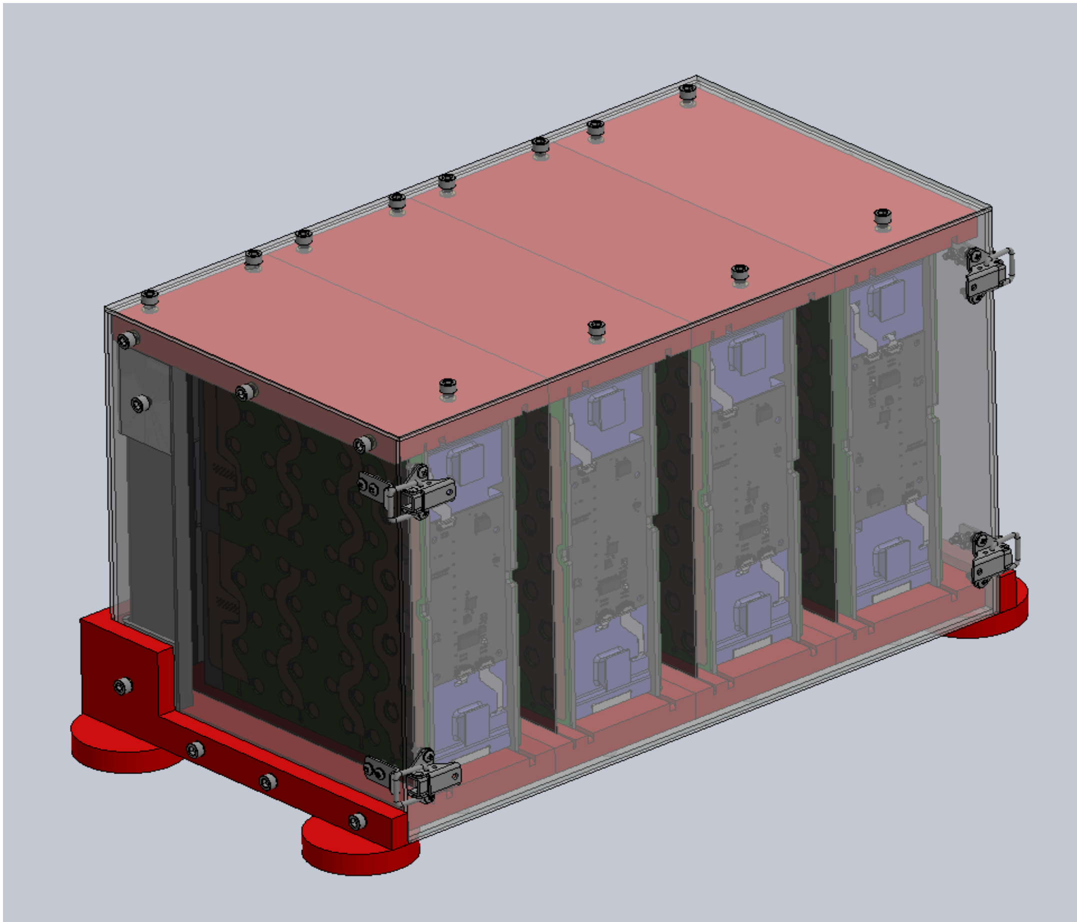
Segment Storage Box



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Abstract

The goal of this independent study project was to create a design and begin manufacturing a storage box for battery segments. Before this project the FSAE team did not have a way to safely store and transport individual segments. This made segments vulnerable when not inside of the accumulator. Each segment holds roughly 60 lithium ion battery cells which can explode when punctured. Having little to no protection for this vital component posed a large safety risk. This storage box is designed to provide the segments with proper protection so this safety risk is eliminated.

The main goal of this independent study project was to finish the design of the storage box and allow for future upgrades due to the team potentially transitioning to different segments next year. The box needs to be able to handle large unexpected forces in case it's dropped or the cabinet it is in gets hit. The following report shows the design and manufacturing process of the segment storage box. This report will go over how the storage box and all of its components were designed and manufactured.

Rails

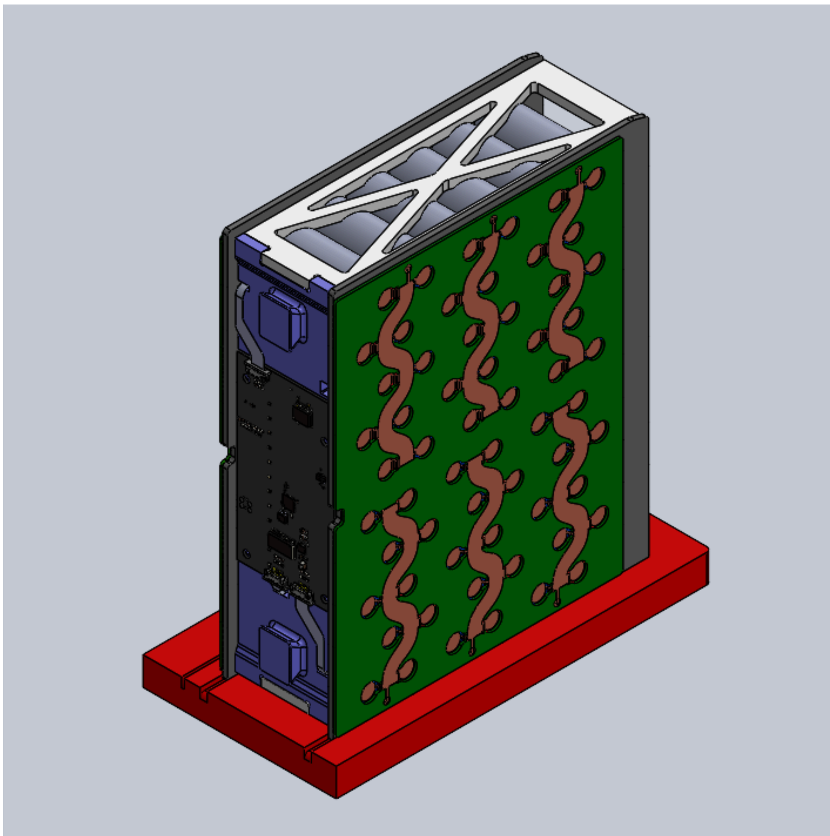

Fig 1: Segment Secured In Rail

The storage box was designed primarily around the rails. The goal of the rails is to keep the segments secure in the box, but at the same time provide little effort from the engineer to slide the segment into the rail. The rails are made out of PETG due to PETG's ease of manufacturing using FDM 3D printing. Another reason why PETG was chosen is due to its material properties, it has a higher tensile strength than ABS and better heat resistance than PLA. The length, width, and thickness of the rails were primarily determined by the size of the segments and the size of Sagamore's flammables cabinet in which the storage box will primarily be stored inside of.

The rails were designed for the intent of an engineer to slide the segments' FR-4 "legs" into the rails' slits allowing for easy entry but resistance on exit so the segments won't slide off and hit the corner of the walls. A slip fit was the target clearance fit because this fit will allow the segments to be secure inside the rails while allowing the engineer to easily slide the segment in. The "legs" of the segments measure 3.2 millimeters in width, so a 4 millimeter hole size for the rails were chosen. This fit also takes into account expansion due to heat. A projected peak expansion of 0.028 millimeters was calculated then verified through a finite element analysis.

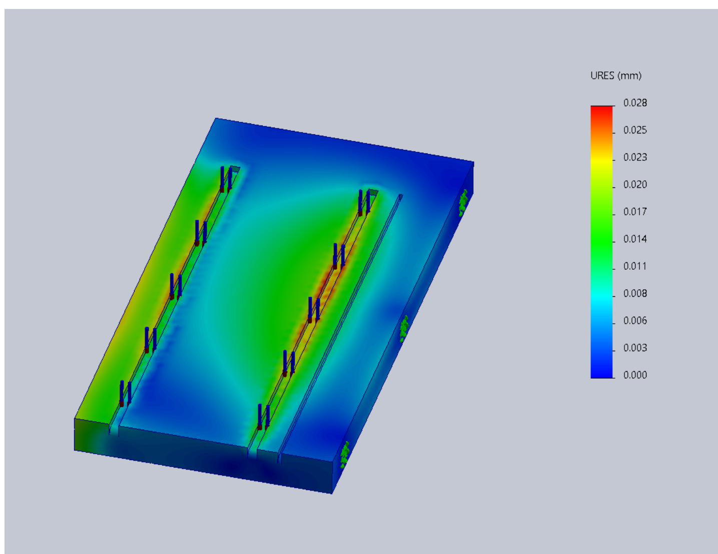

Fig 2: FEA using 60 degree Celsius Load on Rails

The finite element analysis was done using a 60 degree continuous temperature load which was placed on the rails with fixed points placed on the 3 tap holes on the right side and bottom of the rails. A 0.028 millimeter increase still allows the rails to be in range of a slip fit.

As mentioned earlier, PETG was chosen because of its superior heat resistance than PLA. PLA softens at around 55 degrees Celsius while PETG maintains its structure up until 80 degrees Celsius with a continuous load. PETG also has superior tensile strength to ABS (50 MPa vs 30 MPa) which strain is the main stress to account for because of the high heat the rails will experience.

A factor of safety of 4 was the goal due to the team's intention of long term use and because the rails need to observe high amounts of unexpected force in transport or in the flammables cabinet to ensure the segments are protected. The rails will experience 10 MPA of force alone from the temperature load and the weight of the segment. A thickness of 18 millimeters gives us a factor of safety of 4.

FOS: 10 MPA/40 MPA = 4

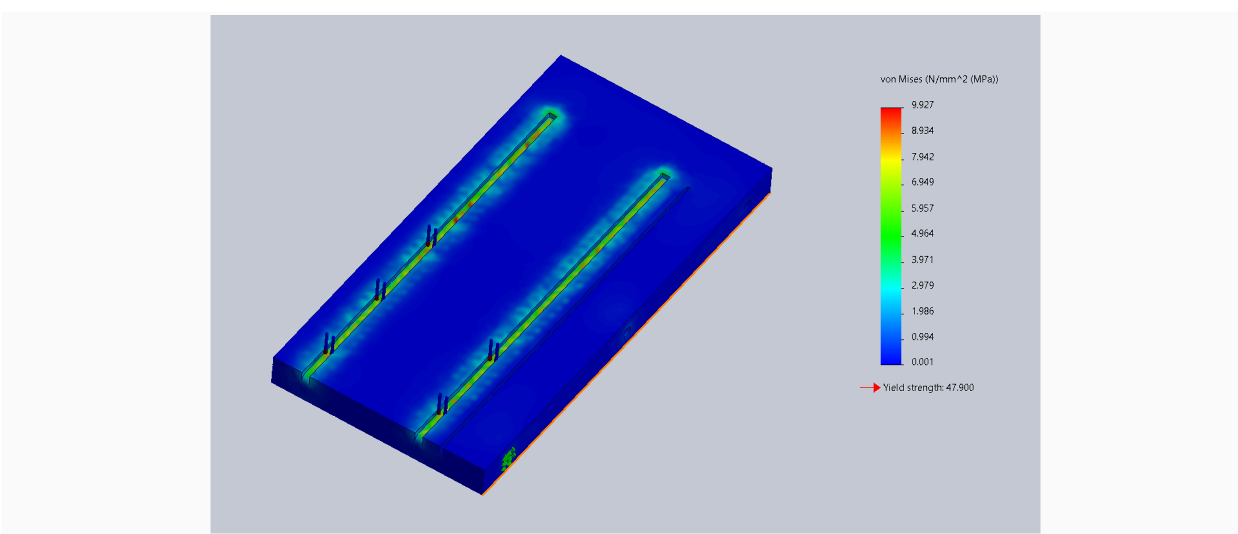

Fig 3: Stress Analysis of Rail

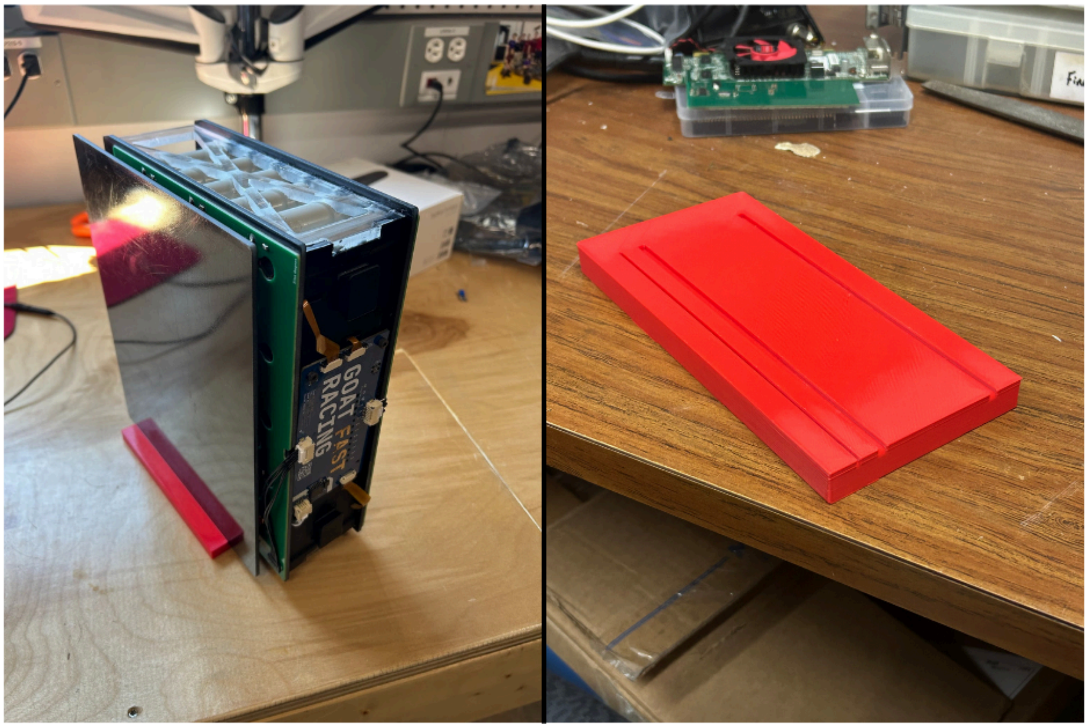

Fig 4: Real Life Prints of The Rails

The left hand side of the figure shows a rail that was printed to test the clearance fit and it was cut to 1/2 the original size to save filament. The right hand side of the figure is a rail printed at full size. The full sized rail measures 238 mm L x 118.15 W x 18 mm H. A few issues occurred when attempting to print the rails. The first issue was that the PETG was warping due to PETG needing a higher temperature to print. So to counter this, the plate temperature was increased to 80 degrees celsius and a 5 millimeter brim was added. The second issue was that shrinkage occurred. Instead of the slits of the rails being 4 millimeters like it was in the CAD, the size would be 3.9 millimeters after printing. This did not affect the overall design and functionality of the rail because the rails were designed with extra tolerance in case this would happen.

A total of 8 (4 on the bottom and 4 on the top) rails are to be produced so that the storage box can hold 4 segments. Each rail is constrained in the box through 12 mm long M6 Stainless Steel Socket Head bolts using 7.62 mm long brass heat inserts. The outermost rails both have 3 bolts (same type) on the side facing the polycarbonate wall, while every rail has 3 bolts on the top or bottom to further constrain the rails to the polycarbonate.

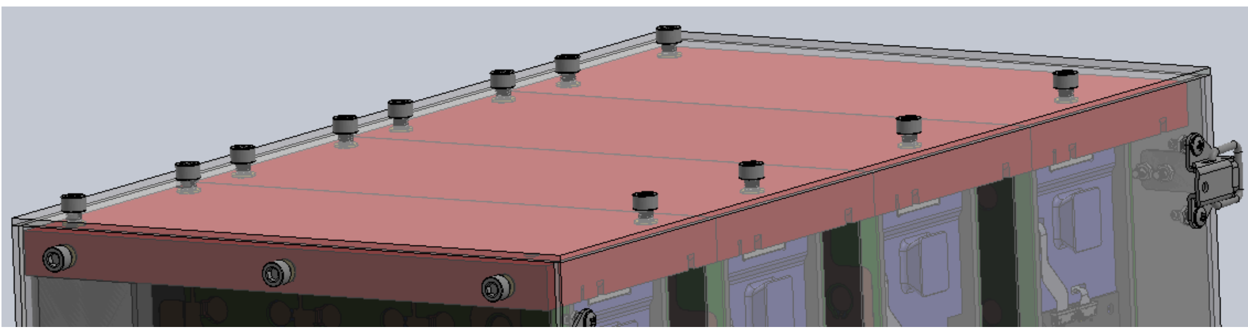

Fig 5: Bolt formation on rails to attach to polycarbonate

Polycarbonate Enclosure

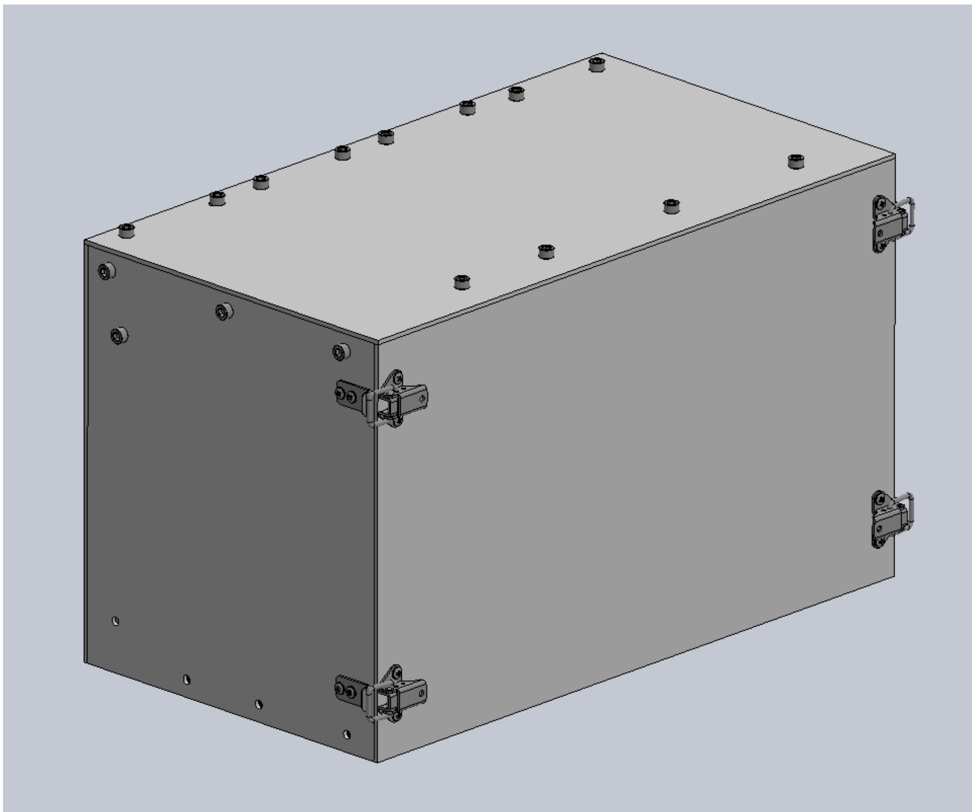

Fig 6: Polycarbonate Enclosure

The enclosure consists of 6 parts: the door, roof, back plate, side walls, and the base plate. Polycarbonate was chosen as the material for the enclosure due to its high strength for a non-metal and because it's an insulator so the segments are not at risk of shorting. The length, width, and height of the polycarbonate was determined based on the rails, segments, and flammables cabinet. The thickness was originally chosen because 0.118 inches was the thinnest polycarbonate could be for SendCutSend to route it.

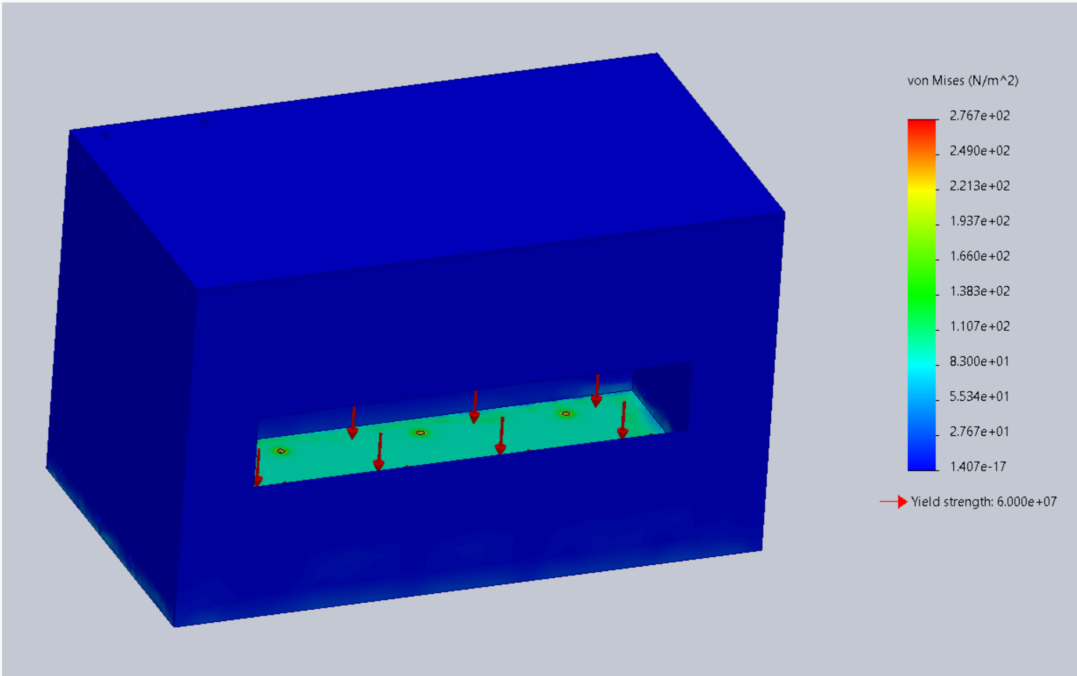

Fig 7: Stress of segments on bottom plate

A finite element analysis confirmed that the 0.118 inch thick polycarbonate could more than handle the 243 Pa force of the segments on the bottom plate. Deflection was also minimal and is not a concern because the box will be lying flat on the bottom plate in the flammable cabinet majority of the time. Corner brackets will be used to prevent bending from the sheets.

Fig 8: Polycarbonate Side Wall Manufactured

Spare polycarbonate that measured 24 inches W x 48 inches L x 0.375 inches H (thickness) was found in Higgins FSAE shop. This large stock was cut in half using a benchtop bandsaw from Washburn, then the side walls were marked on the cut polycarbonate sheet then cut in the bandsaw. Then the side walls were faced down to 0.118 inches using a HAAS VM and a ½ inch facemill with multiple passes to prevent heat build up. The side walls measured 283.1 millimeters L x 255 millimeters W x 3 millimeters H after being cut which is very close to the CAD dimensions. The plan is to machine all components of the polycarbonate enclosure in the same way as the side walls were manufactured. Then all components will be attached via DP8005 epoxy. For all the taps the plan is to use a router to create the holes.

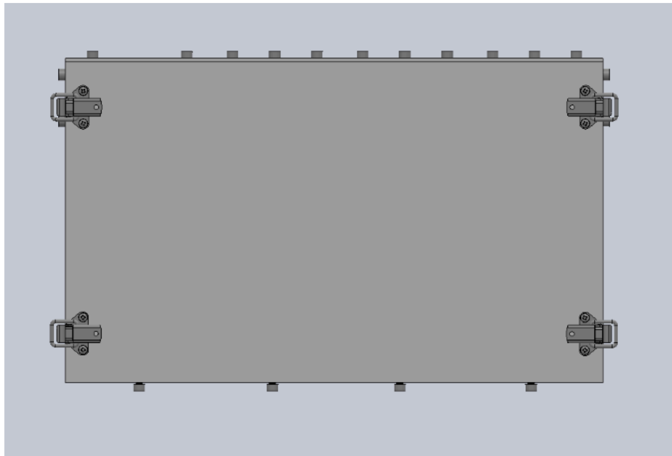

Fig 9: door of the storage box with latches

The polycarbonate door will be bolted on using 4 screw on draw latches. The latches will be connected to the polycarbonate using plastic screws that are secured with plastic nuts. This allows for the door to be removable yet secure. The plastic hardware reduces the risk of shorting the segments in case one of them becomes undone and falls into the storage box.

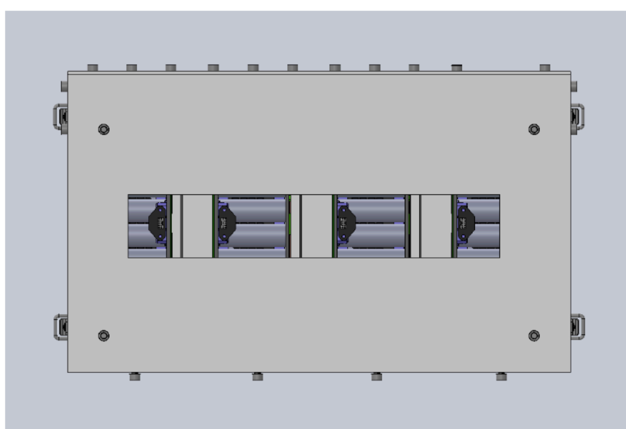

Fig 10: Back plate of storage box

The back plate was given the same dimensions as the door, but it has 4 tap holes to connect aluminum brackets to the corners of the storage box. It was also given a large center rectangle to allow for space and access to the connectors of the segments for charging purposes. This further opens up the potential to charge through the box.

PLA Cover for Corner Brackets

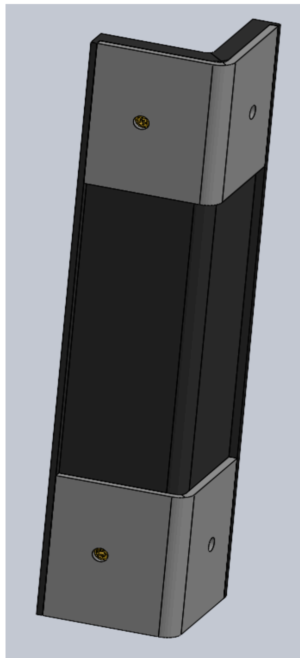

Fig 11: PLA Cover for Aluminum Brackets

4 6061 aluminum corner brackets that are each covered by a PLA cover that has Nomex inside will be used as structural support of the polycarbonate in addition to the epoxy. The aluminum brackets have to be covered to eliminate the risk of the segments shorting. Heat inserts will be used to bolt the PLA cover to the aluminum corner brackets.

TPU Legs

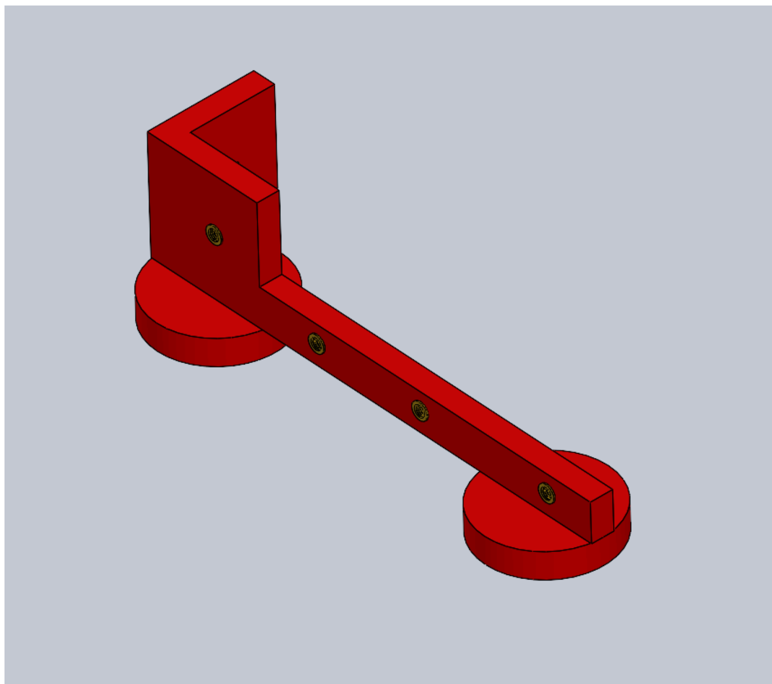

Fig 12: TPU Legs for Storage Box

The storage box contains two "legs" that feature circles (chosen due to higher surface area) to give the box support on the bottom. The legs will be made out of TPU due to its rubber texture and impact absorption capabilities. 5 heat inserts were placed in the holes of the legs to bolt into the polycarbonate to attach the components together.

Conclusion

The storage box's design was finished and some components were manufactured. Plans have been made to continue manufacturing and potentially add upgrades to the storage box's design once next years' segments and electrical systems are planned and produced. The storage box currently allows for such adjustments to be made.